Test 2018, answers

Calculus (University of Toronto)



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MAT137Y1 - Calculus!
Test 1 — May 28th, 2018
Time: 110 minutes

Please, fill this page with ALL CAPITAL LETTERS:

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INSTRUCTIONS: (READ CAREFULLY!)

- **Do not write or draw anything on the QR code on the top corner of any page.**
- This exam booklet contains 13 pages including this one. It consists of 7 questions. The maximum score is 44 points.
- **For questions where you are provided with a box labelled "Your answer:", write your final answer in the box**. Write any justifications, explanations, or calculations you need underneath. **In order to get full marks, you need your final answer to be correct and you need a justification.**
- **Always justify your answers.** An incorrect final answer supported by correct calculations and explanations might still receive partial credit.
- **You can use all the properties and theorems seen in class, unless the question asks explicitly for you to prove something in a particular way.** However, you cannot use any theorem that we have not seen.
- **If you need extra space for a question, you may use Page 13 for this purpose.** If you do so, clearly indicate it on the corresponding problem page.
- **We will only read and grade what you write on the front of each page.** If you need scratch paper, use the back of the pages.
- **Organize your work.** Work that is scattered over the page, that has no clear order, or that is messy and illegible might receive little credit.
- **No aids are permitted on this examination.** Examples of illegal aids include, but are not limited to textbooks, notes, cheatsheets, calculators, cellphones, or any other electronic device.
- **Do not turn this page over until the invigilators instruct you to do so.**

Good luck!

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1. *[2 points each]* Calculate the following four limits, or say they do not exist. For the limits in this question you do not need to provide an $\epsilon - \delta$ proof.

(a) $\lim_{x \to 0} x \sin\left(e^{1/x}\right)$

[redacted]

(b) $\lim_{x \to \pi} \dfrac{\cos^2\left(x^2\right) \tan^2(2x+1)}{\sin^2(2x+1)}$

[redacted]

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(c) $\lim_{x \to 2} \dfrac{\sqrt{3-x}-1}{|2-x|}$

[redacted]

(d) $\lim_{x \to 0} \dfrac{\cos(2x)-1}{\sin(2x)x}$

[redacted]

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2. *[7 points]*

(a) *[2 points]* Write down the $\epsilon - \delta$ definition of the statement

$$\lim_{x \to -1} f(x) = 0.$$

[redacted]

(b) *[5 points]* Let $f(x) = 2|x + 1|$. Use the $\epsilon - \delta$ definition of limit to prove that

$$\lim_{x \to -1} f(x) = 0.$$

[redacted]

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Extra space for Question 2b.

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3. Decide whether each of the following statements is true or false and circle the correct answer. If true, write down a proof. If false, provide a counterexample.

   (a) *[3 points]* If $\lim_{x\to 0} f(|x|)$ exists, then $\lim_{x\to 0} f(x)$ exists.

Answer: (circle one)	**True**	**False**

[redacted]

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(b) *[3 points]* If $\lim_{x \to 0} f(x)$ exists, then $\lim_{x \to 0} f(|x|)$ exists.

Answer: (circle one)	**True**	**False**

[redacted]

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4. Let f and g be functions with domain $\mathbb{R}$. We say that f **encourages** g if for every $x \in \mathbb{R}$, there exists a $y \in [x, x+1]$ such that $g(y) > f(x)$.

   (a) *[2 points]* Write down the definition of "f does NOT encourage g".

   (b) *[3 points]* Give an example, providing explicit equations, of functions f and g with domain $\mathbb{R}$ such that

   $$f \text{ encourages } g \quad \text{AND} \quad g \text{ encourages } f.$$

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5. This question uses the definition of **encourages** from the previous question.

   (a) *[2 points]* State the Intermediate Value Theorem.



   (b) *[4 points]* Let f and g be continuous functions with domain $\mathbb{R}$. Suppose that f encourages g and that

$$\lim_{x \to 1} f(x) = g(0) + 1.$$

   Use the Intermediate Value Theorem to prove that for every $c \in [g(0), g(0) + 1)$, there exists an $x \in \mathbb{R}$ such that $g(x) = c$.



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6. (a) *[3 points]* Let $f(x) = \sqrt{x}$. Use the definition of the derivative to show that $f'(a) = \frac{1}{2\sqrt{a}}$ for all $a > 0$. You do not need to provide an $\epsilon - \delta$ proof here.



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(b) *[4 points]* Let $f(x) = x^{n/2}$ for some positive integer n. Use induction and part (a) to prove that

$$f'(a) = (n/2)a^{n/2-1}$$

for all $a > 0$. For this question you may not use the fact that

$$\frac{d}{dx}(x^r) = rx^{r-1},$$

for values of $r \neq 1/2$ (of course, you are allowed to prove this formula for other any values of r that you require). You are allowed to use general rules for taking derivatives of combinations of functions (i.e. chain rule, product rule, quotient rule).



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7. *[5 points]*

Determine all the values of $k \in \mathbb{R}$ so that the line $y(x) = kx - 3$ is tangent to the graph of the function $f(x) = x^2 - 2x$.



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